MANAGEMENT OPERATING REVIEW - RECORD 004

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-004

Deployment Operating Review for May 2024: purpose and attendance

The packet now carries a single page of exceptions, and everything else is available on request rather than presented. The Deployment Operating Review convened for May 2024. Notes were circulated on 2024-06-06. Redlands became the reference job afterwards. The estimate model had no retrofit class. Hollis Bramwell, Director of Supply Chain, presented the operations section. The next session is set for 2024-07-09. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Owners who could not attend sent written positions, and those were read into the record verbatim. Pre-reads went out ahead of the 2024-06-06 session. One function submitted its pre-read late. Redlands became the reference job afterwards. The estimate model had no retrofit class. Notes were circulated on 2024-06-06. The next session is set for 2024-07-09. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Guests were limited to the sections that concern them, ending a habit of full attendance for its own sake. Dana Whitlock attended as Chief Operating Officer. Notes were circulated on 2024-06-06. Redlands became the reference job afterwards. The estimate model had no retrofit class. One function submitted its pre-read late. Pre-reads went out ahead of the 2024-06-06 session. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

The packet was shortened after a complaint that the previous edition buried the decisions behind twenty pages of charts. Hollis Bramwell, Director of Supply Chain, presented the operations section. Pre-reads went out ahead of the 2024-06-06 session. Redlands became the reference job afterwards. The estimate model had no retrofit class. Dana Whitlock attended as Chief Operating Officer. The Deployment Operating Review convened for May 2024. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Bookings and order intake

Field feedback contradicted the commercial narrative on one account, and the review chose to believe the field. Intake closed at 55 units for May 2024. Win rate held at 29 percent on 88 live opportunities. Redlands became the reference job afterwards. The estimate model had no retrofit class. Tenmile Freight accounted for the largest single order. The book took 55 units, against 142 last period. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

The order book now holds enough signed volume to fill the build plan through the next two build cycles without new intake. The book took 55 units, against 142 last period. Signed volume: 55 units, win rate 29 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Win rate held at 29 percent on 88 live opportunities. Open pursuits number 88, up from 86. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Sales operations rebuilt the stage definitions after a review found three deals sitting in a proof-of-concept stage with no active pilot. Open pursuits number 88, up from 86. Win rate held at 29 percent on 88 live opportunities. Redlands became the reference job afterwards. The estimate model had no retrofit class. Tenmile Freight accounted for the largest single order. Discount depth peaked on the Havering Logistics multi-site award. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Enterprise accounts moved more slowly than the mid-market this period, and the difference sits almost entirely in procurement and security review. Tenmile Freight accounted for the largest single order. Intake closed at 55 units for May 2024. Redlands became the reference job afterwards. The estimate model had no retrofit class. Win rate held at 29 percent on 88 live opportunities. Discount depth peaked on the Havering Logistics multi-site award. Slipped deals 5, including

Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Renewal conversations for the fleet software attached faster than expected, and the attach rate is now high enough to model separately. Discount depth peaked on the Havering Logistics multi-site award. Intake closed at 55 units for May 2024. Redlands became the reference job afterwards. The estimate model had no retrofit class. Win rate held at 29 percent on 88 live opportunities. Signed volume: 55 units, win rate 29 percent. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Bookings quality, measured by how much of the signed volume has a named site and a readiness date, improved without any change in headline volume. Open pursuits number 88, up from 86. Tenmile Freight accounted for the largest single order. Redlands became the reference job afterwards. The estimate model had no retrofit class. Signed volume: 55 units, win rate 29 percent. Win rate held at 29 percent on 88 live opportunities. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Leavers and backfill position, May 2024

Function	Leavers in period	Annualised attrition %	Primary stated reason	Backfill position
Engineering	0	6	manager relationship	contractor covering
Manufacturing	2	14	retirement	deferred to next quarter
Deployment and field service	0	9	performance	requisition open
Commercial	2	16	travel load	backfilled internally
Customer success	1	11	compensation	not backfilling
Supply chain	3	18	relocation	contractor covering
Finance and administration	1	7	career change outside the industry	deferred to next quarter
People	4	15	manager relationship	requisition open
Engineering (contract)	0	10	retirement	backfilled internally
Manufacturing (contract)	3	17	performance	not backfilling
Deployment and field service (contract)	1	8	travel load	contractor covering
Commercial (contract)	3	13	compensation	deferred to next quarter
Customer success (contract)	0	6	relocation	requisition open

Deal inspection

The largest pursuit in the book carries a dependency on the customer finishing a racking project that has already been delayed twice. Two losses went to the same competitor. Hollis Bramwell owns the re-qualification pass. Redlands became the reference job afterwards. The estimate model had no retrofit class. Weighted coverage sits near 88 open pursuits. The Pinebank Distribution pursuit re-dated again. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Procurement cycles at the grocery accounts run longer than the model assumes, and the model has not been updated since it was written. The Pinebank Distribution pursuit re-dated again. Two losses went to the same competitor. Redlands became the reference job afterwards. The estimate model had no retrofit class. Deals leaving the period: 5, most still dated. Hollis Bramwell owns the re-qualification pass. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

One displacement pursuit has consumed more engineering support hours than the next four combined, with no commitment in return. Weighted coverage sits near 88 open pursuits. Conversion into commercial close remains the weak stage. Redlands became the reference job afterwards. The estimate model had no retrofit class. Two losses went to the same competitor. Slippage totalled 5 opportunities against 9 before. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

The team agreed to stop reporting weighted pipeline in this meeting, on the grounds that nobody had ever acted on the number. Conversion into commercial close remains the weak stage. Deals leaving the period: 5, most still dated. Redlands became the reference job afterwards. The estimate model had no retrofit class. Slippage totalled 5 opportunities against 9 before. The Pinebank Distribution pursuit re-dated again. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Of the opportunities that left the quarter, most remain live and dated, and only a minority were lost to a competitor outright. Deals leaving the period: 5, most still dated. Hollis Bramwell owns the re-qualification pass. Redlands became the reference job afterwards. The estimate model had no retrofit class. Weighted coverage sits near 88 open pursuits. Slippage totalled 5 opportunities against 9 before. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Two accounts that were treated as sure things at the last review have gone quiet, and neither seller has escalated. Deals leaving the period: 5, most still dated. Two losses went to the same competitor. Redlands became the reference job afterwards. The estimate model had no retrofit class. Conversion into commercial close remains the weak stage. Weighted coverage sits near 88 open pursuits. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Burn by category, May 2024

Category	Period USD thousands	Prior period USD thousands	Movement %	Commentary
Engineering payroll	112	737	-84.8	qualification testing
Manufacturing payroll and overheads	495	510	-2.9	contract crews on two sites
Deployment labour, employees	391	510	-23.3	spares buy for the drive unit
Deployment labour, contract crews	234	364	-35.7	event spend deferred
Commercial payroll	208	437	-52.4	contract crews on two sites
Marketing and events	182	291	-37.5	contract crews on two sites
Field service payroll	104	364	-71.4	contract crews on two sites
Spares and consumables	156	291	-46.4	second hub fit-out
Facilities and rent	130	218	-40.4	headcount ramp
Information technology and tooling	130	145	-10.3	test cell deferred
Professional services	104	291	-64.3	qualification testing
Recruiting fees	78	145	-46.2	qualification testing
Travel and subsistence	52	218	-76.1	test cell deferred
Insurance and compliance	104	291	-64.3	test cell deferred
Test and validation materials	52	1458	-96.4	event spend deferred
Capital expenditure	78	1020	-92.4	event spend deferred

Site queue and go-live performance

Crew travel days per site came down, largely because two regional hubs now hold enough tooling to avoid shipping kit ahead of each job. Dana Whitlock sequenced the queue by readiness. Contract to go-live averaged 71 days. Redlands became the reference job afterwards. The estimate model had no retrofit class. The queue carries 12 sites, 11 a period ago. Go-live cycle: 71 days, from 59. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Acceptance testing found more integration defects than mechanical ones, and the defects clustered in the warehouse-management adapter. Go-live cycle: 71 days, from 59. Grand Forks and Redlands both cleared acceptance testing. Redlands became the reference job afterwards. The estimate model had no retrofit class. Contract to go-live averaged 71 days. The queue carries 12 sites, 11 a period ago. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Installer utilisation is high enough to be a risk rather than an achievement, because there is no slack for rework travel. Go-live cycle: 71 days, from 59. The queue carries 12 sites, 11 a period ago. Redlands became the reference job afterwards. The estimate model had no retrofit class. Grand Forks and Redlands both cleared acceptance testing. Dana

Whitlock sequenced the queue by readiness. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Every site that slipped this period slipped for a reason on the customer side, which is a better position than last year and still not good enough. The queue carries 12 sites, 11 a period ago. Grand Forks and Redlands both cleared acceptance testing. Redlands became the reference job afterwards. The estimate model had no retrofit class. Contract to go-live averaged 71 days. Go-live cycle: 71 days, from 59. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Sequencing the queue by contractual date rather than by readiness produced two expensive standstills, and the rule has been reversed. Queue depth reached 12 sites. Go-live cycle: 71 days, from 59. Redlands became the reference job afterwards. The estimate model had no retrofit class. Grand Forks and Redlands both cleared acceptance testing. Contract to go-live averaged 71 days. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

A retrofit site behaved nothing like a greenfield site, and the estimate model still treats them as the same class of work. Grand Forks and Redlands both cleared acceptance testing. Contract to go-live averaged 71 days. Redlands became the reference job afterwards. The estimate model had no retrofit class. Queue depth reached 12 sites. Go-live cycle: 71 days, from 59. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Installation capacity

Crew composition changed this period, with more integration skill and less mechanical assembly skill in each team. Crew leads remain the gating resource. Marguerite Do covered two regions this period. Redlands became the reference job afterwards. The estimate model had no retrofit class. Installer utilisation ran at 91 percent. Certified crew count trails the site queue of 12. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Scheduling crews more than six weeks out has proved futile, because customer readiness dates move inside that window. Crew leads remain the gating resource. Utilisation was 91 percent against 90 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Salinas Valley needed a second crew visit. Travel days per site fell again. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Training throughput limits how fast the crew grows, because each certifying lead can supervise one trainee per site without degrading the job. Installer utilisation ran at 91 percent. Crew leads remain the gating resource. Redlands became the reference job afterwards. The estimate model had no retrofit class. Travel days per site fell again. Salinas Valley needed a second crew visit. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

A crew lead shortage in one region was covered by borrowing from another, which worked once and cannot be the plan. Marguerite Do covered two regions this period. Crew leads remain the gating resource. Redlands became the reference job afterwards. The estimate model had no retrofit class. Installer utilisation ran at 91 percent. Certified crew count trails the site queue of 12. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Contractor quality varied by firm rather than by individual, and the sourcing decision now reflects that. Travel days per site fell again. Installer utilisation ran at 91 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Salinas Valley needed a second crew visit. Certified crew count trails the site queue of 12. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

The apprenticeship route produced its first fully certified engineer, which took longer than planned and cost less than lateral hiring. Utilisation was 91 percent against 90 percent. Marguerite Do covered two regions this period. Redlands became the reference job afterwards. The estimate model had no retrofit class. Installer utilisation ran at 91

percent. Salinas Valley needed a second crew visit. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Collections and days outstanding by account, May 2024

Account	Invoiced USD thousands	Received USD thousands	Days outstanding	Collection blocker
Tenmile Freight	432	207	52	customer audit of milestones
Bexley Pharmaceutical Wholesale	360	143	73	credit note pending
Marrowfield Grocers	315	512	58	purchase order number mismatch
Pinebank Distribution	248	369	79	acceptance document unsigned
Alder & Vance Retail	202	328	46	payables portal submission
Northstar Cold Chain	157	266	70	none
Havering Logistics	541	225	55	customer audit of milestones
Quillon Home Supply	216	103	76	credit note pending
Tenmile Freight - second site	180	71	49	purchase order number mismatch
Bexley Pharmaceutical Wholesale - second site	157	256	64	acceptance document unsigned
Marrowfield Grocers - second site	124	184	43	payables portal submission
Pinebank Distribution - second site	101	164	67	none
Alder & Vance Retail - second site	78	133	52	customer audit of milestones
Northstar Cold Chain - second site	270	112	73	credit note pending
Havering Logistics - second site	144	69	58	purchase order number mismatch

Build plan against actual

Rebalancing the two assembly cells lifted output without adding people, mostly by moving the harness dress-up operation upstream. Output reached 49 units, from 45. Plan was 52 units; the line delivered 49. Redlands became the reference job afterwards. The estimate model had no retrofit class. Work in process ended at 29 units. Ingrid Solberg owns the line rebalance. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Cycle time on the pallet mover remains longer than the tote handler by more than the work content justifies. Plan was 52 units; the line delivered 49. Work in process ended at 29 units. Redlands became the reference job afterwards. The estimate model had no retrofit class. Overtime carried the closing two weeks. Attainment gap: 3 units short of plan. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Kitting accuracy improved after the pick-to-light change, and the number of build starts held for missing parts fell accordingly. Overtime carried the closing two weeks. Plan was 52 units; the line delivered 49. Redlands became the reference job afterwards. The estimate model had no retrofit class. Output reached 49 units, from 45. The HS-400 pallet mover drove most of the changeover time. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Configuration variety on the order book has grown faster than the line was designed for, and changeover time shows it. The HS-400 pallet mover drove most of the changeover time. Ingrid Solberg owns the line rebalance. Redlands became the reference job afterwards. The estimate model had no retrofit class. Output reached 49 units, from 45. Overtime carried the closing two weeks. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Build attainment against plan was short again, and the shortfall traces to component availability rather than to line throughput. Work in process ended at 29 units. Plan was 52 units; the line delivered 49. Redlands became the reference job afterwards. The estimate model had no retrofit class. The HS-400 pallet mover drove most of the changeover time.

Attainment gap: 3 units short of plan. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Overtime carried the last two weeks of the period, which closed the gap on units and did nothing good for cost per unit. Attainment gap: 3 units short of plan. The HS-400 pallet mover drove most of the changeover time. Redlands became the reference job afterwards. The estimate model had no retrofit class. Work in process ended at 29 units. Overtime carried the closing two weeks. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Carried constraints

Constraint under review: field engineering capacity, at the capacity math stage as of this session. Installer utilisation sat at ninety-one percent across the crew, which leaves no absorption for rework travel, and two sites in the queue have no named crew lead at all. Bookings 55 units across 88 pursuits. Pack and ship ran 94 percent first pass.

Constraint under review: large-account uptime, at the baseline stage as of this session. Fleet uptime at Cedar Bluff has run at ninety-six percent against a contractual floor of ninety-seven, and the account team has not yet had to write a service credit. On-time delivery 83 percent, Tessellate Controls the outlier. Work in process 29 units, mostly Castings.

Dual sourcing costs more per unit during the overlap period, and the review accepted that cost with its eyes open. Supplier on-time delivery came in at 83 percent. Hollis Bramwell issued the first supplier scorecards. Redlands became the reference job afterwards. The estimate model had no retrofit class. Coverage weeks on long-lead parts were extended. Ironbark Sheet Metal missed two consecutive commit dates. Win rate 29 percent, best in Ontario. Queue depth 12 sites, Grand Forks at the front. Days outstanding 61, Tenmile Freight the largest balance. Work in process 29 units, mostly Castings.

Runway scenarios tabled in the May 2024 review

Scenario	Horizon	Monthly burn USD thousands	Hiring posture	Runway months	Arming trigger
Upside	next quarter	2820	plan plus eight roles	21	two multi-site awards signed by quarter end
Upside	two quarters out	2960	plan plus eight roles	20	two multi-site awards signed by quarter end
Upside	four quarters out	3140	plan plus eight roles	19	two multi-site awards signed by quarter end
Middle, planning case	next quarter	2610	plan as approved	23	bookings within ten percent of plan
Middle, planning case	two quarters out	2750	plan as approved	22	bookings within ten percent of plan
Middle, planning case	four quarters out	2930	plan as approved	21	bookings within ten percent of plan
Downside	next quarter	2430	hiring pause on internal roles	26	two large awards leave the quarter
Downside	two quarters out	2570	hiring pause on internal roles	25	two large awards leave the quarter
Downside	four quarters out	2750	hiring pause on internal roles	24	two large awards leave the quarter
Severe	next quarter	2270	pause plus contract labour reduction	29	site queue shorter than nine sites
Severe	two quarters out	2410	pause plus contract labour reduction	28	site queue shorter than nine sites
Severe	four quarters out	2590	pause plus contract labour reduction	27	site queue shorter than nine sites

Manufacturing quality

Escapes to the field are the measure that matters, and this period produced a number that the service team could absorb. Scrap totalled 24 thousand dollars. Scrap ran 24 thousand, from 26 thousand. Redlands became the reference job afterwards. The estimate model had no retrofit class. Yield was 94 percent against 93 percent. Rework hours concentrated in the sensor cell. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

A supplier process change arrived without notification and produced a yield loss that took five weeks to attribute correctly. Tessellate Controls changed a process without notice. First-pass yield finished at 94 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Yield was 94 percent against 93 percent. Scrap ran 24 thousand, from 26 thousand. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Warranty claims lag the build date by months, so the yield excursion will keep appearing in service data after the plant has moved on. First-pass yield finished at 94 percent. Yield was 94 percent against 93 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Owen Castellanos signed the corrective action. Tessellate Controls changed a process without notice. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

First-pass yield is the metric that moved most this period, and the movement is confined to one cell rather than spread across the line. Scrap totalled 24 thousand dollars. Owen Castellanos signed the corrective action. Redlands became the reference job afterwards. The estimate model had no retrofit class. Scrap ran 24 thousand, from 26 thousand. Yield was 94 percent against 93 percent. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Root cause work took longer than the containment did, which is the correct order and still frustrating to watch. Owen Castellanos signed the corrective action. First-pass yield finished at 94 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Tessellate Controls changed a process without notice. Scrap totalled 24 thousand dollars. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Rework hours in the sensor cell exceeded the assembly hours for two weeks, which is the clearest signal the process was out of control. Scrap ran 24 thousand, from 26 thousand. Rework hours concentrated in the sensor cell. Redlands became the reference job afterwards. The estimate model had no retrofit class. Scrap totalled 24 thousand dollars. Yield was 94 percent against 93 percent. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Supply base performance

Purchase order acknowledgements are now chased on a fixed cadence, which sounds trivial and caught two silent date pushes. On-time delivery was 83 percent, from 88 percent. Supplier on-time delivery came in at 83 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Penrose Castings improved after a site visit. Ironbark Sheet Metal missed two consecutive commit dates. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Forecast sharing with the top five suppliers began this period, and two of them immediately improved their commit dates. Expedite activity rose with the build plan of 52. Coverage weeks on long-lead parts were extended. Redlands became the reference job afterwards. The estimate model had no retrofit class. Penrose Castings improved after a site visit. On-time delivery was 83 percent, from 88 percent. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Dual sourcing costs more per unit during the overlap period, and the review accepted that cost with its eyes open. Coverage weeks on long-lead parts were extended. Ironbark Sheet Metal missed two consecutive commit dates. Redlands became the reference job afterwards. The estimate model had no retrofit class. Supplier on-time delivery came in at 83 percent. Expedite activity rose with the build plan of 52. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Tooling ownership is unclear on four part numbers, and the review asked for the contracts to be read rather than remembered. Supplier on-time delivery came in at 83 percent. Penrose Castings improved after a site visit. Redlands became the reference job afterwards. The estimate model had no retrofit class. Ironbark Sheet Metal missed two consecutive commit dates. Hollis Bramwell issued the first supplier scorecards. Slipped deals 5, including Havering

Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Where tooling belongs to the supplier, the exit cost is real and it is not written down anywhere. Penrose Castings improved after a site visit. On-time delivery was 83 percent, from 88 percent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Hollis Bramwell issued the first supplier scorecards. Ironbark Sheet Metal missed two consecutive commit dates. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Lead-time creep is the pattern to watch, because quoted lead times moved without any supplier formally notifying a change. On-time delivery was 83 percent, from 88 percent. Coverage weeks on long-lead parts were extended. Redlands became the reference job afterwards. The estimate model had no retrofit class. Ironbark Sheet Metal missed two consecutive commit dates. Penrose Castings improved after a site visit. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Roadmap checkpoints against customer commitments, May 2024

Checkpoint	Product line	Committed date	Status	Dependency
Fleet orchestration release candidate	HS-200 tote handler	2024-06-27	stopped by decision	customer test window
Warehouse management adapter for the grocery host	HS-400 pallet mover	2024-07-19	slipped one milestone	field data volume
Docking regression fix	Orchestrator fleet software	2024-08-10	stopped by decision	none
Controller revision B	Beacon WMS adapter	2024-09-01	slipped one milestone	none
Battery health telemetry	HS-200 tote handler	2024-09-23	stopped by decision	field data volume
Simulation coverage target	HS-400 pallet mover	2024-10-15	slipped one milestone	test cell availability
Operator training module	Orchestrator fleet software	2024-11-06	stopped by decision	customer test window
Navigation parameter standardisation	Beacon WMS adapter	2024-11-28	slipped one milestone	supplier firmware
Fleet orchestration release candidate - second increment	HS-200 tote handler	2024-12-20	stopped by decision	supplier firmware
Warehouse management adapter for the grocery host - second increment	HS-400 pallet mover	2025-01-11	slipped one milestone	certification of crew leads
Docking regression fix - second increment	Orchestrator fleet software	2025-02-02	stopped by decision	field data volume
Controller revision B - second increment	Beacon WMS adapter	2025-02-24	slipped one milestone	certification of crew leads
Battery health telemetry - second increment	HS-200 tote handler	2025-03-18	stopped by decision	none
Simulation coverage target - second increment	HS-400 pallet mover	2025-04-09	slipped one milestone	supplier firmware
Operator training module - second increment	Orchestrator fleet software	2025-05-01	stopped by decision	field data volume
Navigation parameter standardisation - second increment	Beacon WMS adapter	2025-05-23	slipped one milestone	field data volume

Cash burn by category

Every scenario assumes the deployment cost per site keeps falling, and that assumption deserves more scrutiny than it has had. Burn for the period was 2610 thousand dollars. Hollis Bramwell holds the trigger list. Redlands became the reference job afterwards. The estimate model had no retrofit class. Test-cell capital was deferred again. Runway stands at 23 months on the middle case. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Fixed commitments have grown to the point where the fastest lever left is contract labour. Capital spend was 340 thousand dollars. Spending ran 2610 thousand against 7290 thousand. Redlands became the reference job afterwards. The estimate model had no retrofit class. Test-cell capital was deferred again. Hollis Bramwell holds the trigger list. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Runway under the middle scenario extended, and the review chose to plan against that case while holding triggers for the downside. Hollis Bramwell holds the trigger list. Burn for the period was 2610 thousand dollars. Redlands became the reference job afterwards. The estimate model had no retrofit class. Runway stands at 23 months on the middle case. Spending ran 2610 thousand against 7290 thousand. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Scenario planning is only useful if the triggers are specific, so each scenario now names the observable that fires it. Capital spend was 340 thousand dollars. Runway stands at 23 months on the middle case. Redlands became the reference job afterwards. The estimate model had no retrofit class. Test-cell capital was deferred again. Runway moved from 24 to 23 months. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Gross margin per deployed unit is improving slowly, and the improvement is in installation labour rather than in bill of materials. Runway moved from 24 to 23 months. Test-cell capital was deferred again. Redlands became the reference job afterwards. The estimate model had no retrofit class. Capital spend was 340 thousand dollars. Burn for the period was 2610 thousand dollars. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

The finance team rebuilt the model bottom-up from the site queue, and the new version disagrees with the old one on timing rather than on total. Runway moved from 24 to 23 months. Hollis Bramwell holds the trigger list. Redlands became the reference job afterwards. The estimate model had no retrofit class. Spending ran 2610 thousand against 7290 thousand. Capital spend was 340 thousand dollars. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Attendance and pre-read submission, May 2024

Attendee	Function	Section owned	Pre-read submitted
Ingrid Solberg	Chief Financial Officer	cash and runway	yes
Tobias Nkemdirim	VP Engineering	people	late
Renata Voss	VP People	product checkpoints	yes
Hollis Bramwell	Director of Supply Chain	none, attending for context	yes
Sunil Achterberg	Director of Field Service	bookings and pipeline	yes
Marguerite Do	Director of Sales Operations	manufacturing and supply	yes
Emil Farkas	Director of Manufacturing Engineering	deployment operations	late
Nadia Oyelaran	Chief of Staff	field reliability	yes
Griffin Tasse	Head of Product	cash and runway	yes
Lucia Mbeki-Hart	Controller	people	yes
Dana Whitlock	Chief Operating Officer	product checkpoints	yes
Marcus Peil	VP Operations	none, attending for context	late
Priya Ramanathan	VP Deployment	bookings and pipeline	yes
Owen Castellanos	VP Customer Success	manufacturing and supply	yes
Site lead, Cedar Bluff	Customer success, by video	none, attending for context	not required
Crew lead, southern region	Deployment, by video	none, attending for context	not required
Controller's analyst	Finance	none, attending for context	not required
Plant supervisor, second shift	Manufacturing	none, attending for context	not required

Roadmap against commitments

Committed customer dates and internal roadmap dates were put on the same page for the first time, and they did not match. Adapter work crowded out platform work again. Two roadmap checkpoints cleared; one slipped. Redlands became the reference job afterwards. The estimate model had no retrofit class. Field data now sets the reliability backlog order. Documentation lags the current release. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

Release quality improved once simulation coverage became a gate rather than a report. Field data now sets the reliability backlog order. Adapter work crowded out platform work again. Redlands became the reference job afterwards. The estimate model had no retrofit class. Dana Whitlock brought the decision to stop one feature. Two roadmap checkpoints cleared; one slipped. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Fleet orchestration release slipped a milestone, and the slip was accepted in exchange for keeping the docking regression fix on schedule. The HS-200 tote handler absorbed most of the engineering effort. Adapter work crowded out platform work again. Redlands became the reference job afterwards. The estimate model had no retrofit class. Documentation lags the current release. Field data now sets the reliability backlog order. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Field data now drives the reliability backlog, which is a change from prioritising by the loudest account. Dana Whitlock brought the decision to stop one feature. Adapter work crowded out platform work again. Redlands became the reference job afterwards. The estimate model had no retrofit class. Documentation lags the current release. Field data now sets the reliability backlog order. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Pilot features running at two accounts have no path to general availability, and that gap was made explicit. Dana Whitlock brought the decision to stop one feature. Release staging across sites is now standard. Redlands became the reference job afterwards. The estimate model had no retrofit class. Two roadmap checkpoints cleared; one slipped. Adapter work crowded out platform work again. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

Engineering capacity allocation between new sites and installed base was reset toward the installed base this period. The HS-200 tote handler absorbed most of the engineering effort. Two roadmap checkpoints cleared; one slipped. Redlands became the reference job afterwards. The estimate model had no retrofit class. Documentation lags the current release. Adapter work crowded out platform work again. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Recorded disagreement

A request to revisit the sequencing rule was granted, with a date and a named piece of evidence attached. The objection was noted against the decision. No consensus was reached; the VP Deployment decided. Redlands became the reference job afterwards. The estimate model had no retrofit class. The minority view is preserved in the record. The disagreement is recorded rather than resolved. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

The engineering view is that the deferral saves less than it costs, and that view is noted in full. A further test was added because of the dissent. The minority view is preserved in the record. Redlands became the reference job afterwards. The estimate model had no retrofit class. No consensus was reached; the VP Deployment decided. The objection was noted against the decision. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

There was no consensus on the hiring pause, and the pause proceeded with the dissent recorded against it. The disagreement is recorded rather than resolved. Hollis Bramwell and Dana Whitlock read the data differently. Redlands became the reference job afterwards. The estimate model had no retrofit class. A further test was added because of the dissent. Owen Castellanos asked for the decision to be revisited. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

The customer success position, that credits should be offered before they are claimed, did not carry the room. Owen Castellanos asked for the decision to be revisited. Hollis Bramwell and Dana Whitlock read the data differently. Redlands became the reference job afterwards. The estimate model had no retrofit class. The objection was noted against the decision. A further test was added because of the dissent. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Objection was raised to closing an item without measured evidence, and the item stayed open as a result. No consensus was reached; the VP Deployment decided. The minority view is preserved in the record. Redlands became the reference job afterwards. The estimate model had no retrofit class. Owen Castellanos asked for the decision to be revisited. A further test was added because of the dissent. Go-live cycle 71 days at Redlands. Pack and ship ran 94 percent first pass. Ingrid Solberg and Owen Castellanos disagreed on sequencing. Attrition 12 percent annualised in Ontario.

The stricter uptime definition won on the argument that a customer will use it whether we do or not. The objection was noted against the decision. A further test was added because of the dissent. Redlands became the reference job afterwards. The estimate model had no retrofit class. Owen Castellanos asked for the decision to be revisited. Hollis Bramwell and Dana Whitlock read the data differently. Queue depth 12 sites, Grand Forks at the front. Capital spend 340 thousand this period. Headcount 129 of 138, Hollis Bramwell reporting. Depot fill 87 percent at Atlanta depot.

Items deferred from the May 2024 agenda

Item	Reason for deferral	Revisit at	Requested by
Complete process capability study on the bond operation	customer conversation not yet held	2024-08-05	Emil Farkas
Decide disposition of the superseded sensor head stock	wants a second quarter of evidence	2024-07-09	Marcus Peil
Bring deal-level notes for the top ten pursuits	data not good enough to decide	2024-09-04	Marguerite Do
Measure onboarding time to productivity for crews	owner absent from the session	2024-07-09	Renata Voss
Split the reliability backlog by build vintage	commits capital beyond the delegated threshold	2024-10-04	Tobias Nkemdirim
Read the tooling ownership clauses on four part numbers	depends on an open qualification result	2024-07-09	Hollis Bramwell
Re-cut the stage definitions in the pipeline report	customer conversation not yet held	2024-11-03	Marguerite Do
Move collections calls to the account teams	wants a second quarter of evidence	2024-07-09	Lucia Mbeki-Hart
Reconcile the two uptime definitions with the customer	data not good enough to decide	2024-12-03	Owen Castellanos
Rebuild the site cost model separating retrofit from greenfield	owner absent from the session	2024-07-09	Marcus Peil
Draft deposit terms for multi-site commitments	commits capital beyond the delegated threshold	2025-01-02	Ingrid Solberg
Cost the second-shift ramp against overtime	depends on an open qualification result	2024-07-09	Marcus Peil
Reissue the deployment playbook network section	customer conversation not yet held	2025-02-01	Priya Ramanathan
Set a plan freeze window inside the build month	wants a second quarter of evidence	2024-07-09	Emil Farkas

Decisions, owners and dates recorded in May 2024

Nothing was recorded as decided unless somebody in the room owned it and a date was attached to it. Owners were named and dates set against 2024-07-09. Three items carried forward into 2024-07-09. Redlands became the reference job afterwards. The estimate model had no retrofit class. Two due dates were changed in the room. The oldest item dates from two sessions back. Yield 94 percent, scrap 24 thousand in Pack and ship. Days outstanding 61, Tenmile Freight the largest balance. Burn 2610 thousand, runway 23 months. Capital spend 340 thousand this period.

An item was killed rather than rolled a fourth time, and the reasoning is in the record. Marguerite Do, Director of Sales Operations, took the largest open item. Owners were named and dates set against 2024-07-09. Redlands became the reference job afterwards. The estimate model had no retrofit class. The oldest item dates from two sessions back. Three items carried forward into 2024-07-09. Build plan 52, actual 49, Castings short. Utilisation 91 percent, Crew Gulf fully committed. Work in process 29 units, mostly Castings. Burn 2610 thousand, runway 23 months.

Decisions were recorded with a named owner and a date, and items without both were explicitly not recorded as decisions. Two due dates were changed in the room. Marguerite Do, Director of Sales Operations, took the largest open item. Redlands became the reference job afterwards. The estimate model had no retrofit class. Owners were named and dates set against 2024-07-09. Ingrid Solberg inherited an item after a role change. Yield 94 percent, scrap 24 thousand in Pack and ship. Marguerite Do owns the Castings recovery. The HS-200 tote handler took most of the test cell time. Burn 2610 thousand, runway 23 months.

Owners were asked to bring evidence of closure rather than a status colour, and two items failed that test. Three items carried forward into 2024-07-09. Two due dates were changed in the room. Redlands became the reference job afterwards. The estimate model had no retrofit class. Ingrid Solberg inherited an item after a role change. The oldest item

dates from two sessions back. Slipped deals 5, including Havering Logistics. Ironbark Sheet Metal quoted lead times moved again. Win rate 29 percent, best in Ontario. Capital spend 340 thousand this period.

Action items reviewed and recorded, May 2024

Item	Description	Owner	Opened	Due	Status	Closing evidence
OR-0307	Raise geographic radius on field engineering requisitions	Renata Voss	2024-04-25	2024-07-06	Open, date revised	partial
OR-0409	Agree joint remediation plan with the Cedar Bluff account	Owen Castellanos	2024-06-06	2024-07-11	Open	none yet
OR-0420	Publish the supplier scorecard with weighted criteria	Hollis Bramwell	2024-05-28	2024-06-17	Closed	memo circulated
OR-0423	Measure onboarding time to productivity for crews	Renata Voss	2024-05-24	2024-06-23	Open	outstanding
OR-0426	Decide disposition of the superseded sensor head stock	Marcus Peil	2024-05-20	2024-06-29	In progress	outstanding
OR-0429	Stage software releases across sites rather than at once	Griffin Tasse	2024-05-16	2024-07-05	Closed	measured result attached
OR-0432	Read the tooling ownership clauses on four part numbers	Hollis Bramwell	2024-05-12	2024-07-11	Open	outstanding
OR-0435	Reconcile the two uptime definitions with the customer	Owen Castellanos	2024-05-08	2024-07-17	In progress	outstanding
OR-0438	Split the reliability backlog by build vintage	Tobias Nkemdirim	2024-05-04	2024-07-23	Closed	memo circulated
OR-0441	Cost the second-shift ramp against overtime	Marcus Peil	2024-04-30	2024-07-29	Open	outstanding
OR-0444	Bring deal-level notes for the top ten pursuits	Marguerite Do	2024-04-26	2024-08-04	In progress	outstanding
OR-0447	Re-cut the stage definitions in the pipeline report	Marguerite Do	2024-04-22	2024-08-10	Closed	measured result attached
OR-0450	Set a plan freeze window inside the build month	Emil Farkas	2024-04-18	2024-08-16	Open	outstanding
OR-0453	Size the buffer stock on long-lead commodities	Hollis Bramwell	2024-04-14	2024-08-22	In progress	outstanding
OR-0456	Refresh compensation bands for field engineering	Renata Voss	2024-04-10	2024-08-28	Closed	report filed
OR-0459	Complete process capability study on the bond operation	Emil Farkas	2024-04-06	2024-09-03	Open	outstanding